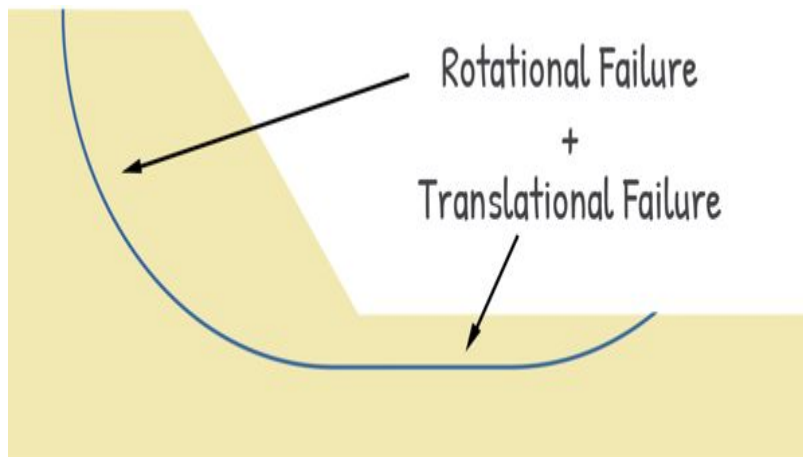




iv. Wedge failure:

- In wedge failure, triangular wedge fails down making angular failure surface.
- Causes: occurs due to adverse geological and hydrological conditions, toe excavation, vibrations, soil erosion, weathering and so on.



2. Introduce bearing capacity of soil. What are the effects of ground water table in the bearing capacity of different types of soil? [2+3=5]

Answer:

Bearing capacity of soil is defined as maximum load per unit area that soil mass can safely withstand without undergoing shear failure or excessive settlement.

It is the ability of a soil mass to safely support structural loads without any failure.

There are two types of bearing capacity of soil. They are:

i. Ultimate bearing capacity:

It is the gross pressure intensity at the base of footing at which soil fails by shear.

The ultimate bearing capacity is commonly expressed by **Terzaghi's bearing capacity equation:**

$$q_u = cN_c + \gamma D_f N_q + 0.5\gamma B N_\gamma$$

Where:

q_u = Ultimate bearing capacity

c = Cohesion of soil

γ = Unit weight of soil

D_f = Depth of foundation

B = Width of foundation

N_c, N_q, N_γ = Bearing capacity factors

ii. Safe bearing capacity:

Maximum pressure which the soil carry safely without risk of shear failure is known as safe bearing capacity. **Safe bearing capacity** is obtained by dividing the ultimate bearing capacity by a suitable factor of safety:

$$q_{safe} = \frac{q_u}{FOS}$$

where **FOS** is the factor of safety.

The effects of ground water table in the bearing capacity of different types of soil:

Groundwater table makes the soil mass submerged. So, unit weight of the soil changes into submerged unit weight.

$$\gamma' = \frac{\gamma}{2}$$

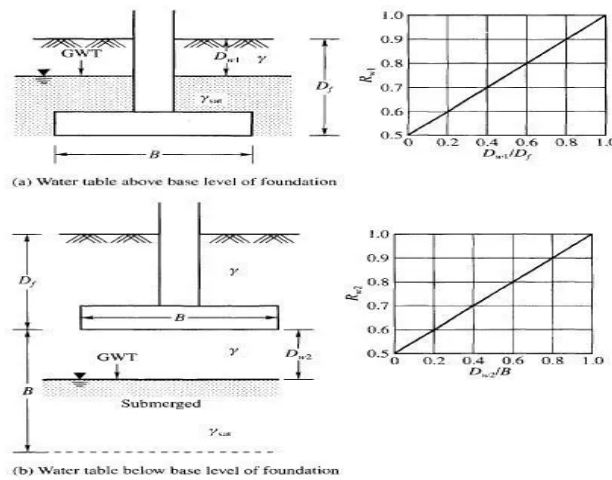


Figure 12.10 Effect of WT on bearing capacity

When Groundwater Table is at Foundation Level:

The soil below the foundation becomes submerged. Therefore, replace γ in the third term with the **submerged unit weight γ'** :

$$\gamma' = \gamma_{sat} - \gamma_w$$

Thus,

$$q_u = cN_c + \gamma D_f N_q + 0.5\gamma' B N_\gamma$$

When Groundwater Table is at Ground Surface:

The soil above and below the foundation is submerged. Use γ' in both second and third terms:

$$q_u = cN_c + \gamma' D_f N_q + 0.5\gamma' B N_\gamma$$

Since: $\gamma' = \gamma_{sat} - \gamma_w < \gamma$, a rise in the groundwater table decreases the effective unit weight of soil, causing the **ultimate bearing capacity q_u to decrease**.

i. For cohesionless soil (sand):

For sandy soils, $c = 0$, so Terzaghi's equation becomes:

$$q_u = \gamma D_f N_q + 0.5\gamma B N_\gamma$$

When the groundwater table rises, the unit weight changes from γ to submerged unit weight γ' :

$$\gamma' = \gamma_{sat} - \gamma_w$$

Therefore,

$$q_u = \gamma' D_f N_q + 0.5\gamma' B N_\gamma$$

Since $\gamma' < \gamma$, $q_u \downarrow$

Effect: Significant reduction in bearing capacity.

ii. **For cohesive soil (clay):**

For clay, Terzaghi's equation becomes:

$$q_u = cN_c + \gamma D_f$$

Since $N_c = 5.7$ and $N_q = 1$,

$$q_u = 5.7c + \gamma D_f$$

The major component is cohesion (c). Therefore, a rise in groundwater affects bearing capacity only slightly in the short term.

$$q_u \approx 5.7c$$

Effect: Small reduction in bearing capacity compared to sand.

iii. **For mixed soil (Silty Clay, Clayey Sand)**

$$q_u = cN_c + \gamma D_f N_q + 0.5\gamma B N_\gamma$$

When the water table rises:

$$q_u = cN_c + \gamma' D_f N_q + 0.5\gamma' B N_\gamma$$

Both cohesion and friction contribute to strength.

Effect: Moderate reduction in bearing capacity.

Mixed soil works better for bearing capacity of soil.

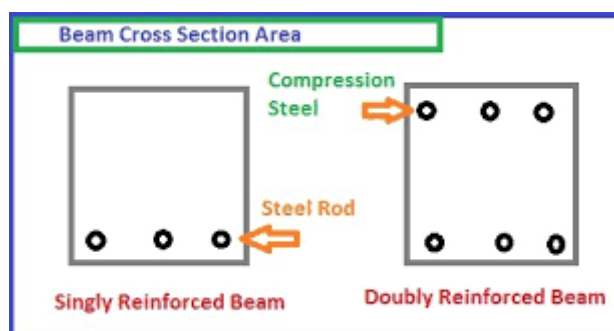
3. **Define singly reinforced and doubly reinforced rectangular concrete sections. Where and when are they used in construction works? Derive an equation for moment of resistance for a balanced section.**

[2+3+5=10]

Answer:

Singly reinforced and doubly reinforced are the two methods of configuration of steel bars in RCC sections. Singly reinforced concrete section can be defined as the concrete section in which the main reinforcement bars are provided only in the tension zone (bottom part of beam section) as the concrete is weak in tension and prone to cracks and failures.

Doubly reinforced concrete sections can be defined as those concrete sections in which the longitudinal steel bars are provided in both tension and compression zone (both top and bottom of the beam sections). Doubly reinforced section is provided when the heavy loads are to be subjected on them without increasing the depth of beam section due to structural and architectural constraint.



Singly reinforced concrete sections are used in construction works:

- when the Bending Moment is relatively small.
- Suitable when the depth of beam is not restricted.

- Commonly used in floor beams, slabs, beams, lintels and other ordinary structural members.

Doubly reinforced concrete sections are used in construction works:

- when the depth of beam is restricted and additional moment capacity is required.
- when the beam is subjected to reversal of stress such as earthquakes.
- to increase stiffness and reduce long term deflection.
- in heavily loaded structures such as bridges, multistorey buildings and retaining structures.

A **balanced section** is the one in which the maximum stress reaches in the concrete as same time as that of steel.

For a rectangular section,

$$C = \frac{1}{2} \sigma_{cbc} b x$$

where:

C = Compression force in concrete

σ_{cbc} = Permissible compressive stress in concrete

b = Width of beam

x = Depth of neutral axis

Tension Force in Steel:

$$T = \sigma_{st} A_{st}$$

where:

T = Tension force in steel

σ_{st} = Permissible tensile stress in steel

A_{st} = Area of tension reinforcement

For a balanced section,

$$C = T$$

Therefore,

$$\frac{1}{2} \sigma_{cbc} b x = \sigma_{st} A_{st}$$

The resultant compression acts at a distance $x/3$ from the top.

Hence the lever arm is

$$z = d - \frac{x}{3}$$

where d is the effective depth.

Moment of Resistance:

$$M_r = C \times z$$

Substituting the values of C and z ,

$$M_r = \left(\frac{1}{2} \sigma_{cbc} b x \right) \left(d - \frac{x}{3} \right)$$

This is the **general expression for the moment of resistance of a balanced section.**

In Terms of Neutral Axis Ratio

Let

$$k = \frac{x}{d}$$

Then,

$$M_r = \frac{1}{2} \sigma_{cbc} b (kd) \left(d - \frac{kd}{3} \right)$$

$$M_r = \frac{1}{2} \sigma_{cbc} b d^2 k \left(1 - \frac{k}{3}\right)$$

$$M_r = Q b d^2$$

where,

$$Q = \frac{1}{2} \sigma_{cbc} k \left(1 - \frac{k}{3}\right) \text{ is called the } \mathbf{moment\ of\ resistance\ factor}.$$

4. The undisturbed soil at a borrow pit has a water content of 15 percent, void ratio of 0.60 and specific gravity of solids 2.70. The soil from the borrow pit is to be used for the construction of an embankment with a finished volume of 40,000 cum. The specifications for the embankment require a water content of 18% and a dry unit weight of 1.76 g/cc. Calculate the quantity of soil required to excavated from the borrow pit for the construction of embankment. [10]

Answer:

Given

Borrow Pit Soil:

$$\text{Water content, } w_1 = 15\% = 0.15$$

$$\text{Void ratio, } e_1 = 0.60$$

$$\text{Specific gravity, } G = 2.70$$

Embankment:

$$\text{Finished volume, } V_e = 40,000 \text{ m}^3$$

$$\text{Water content, } w_2 = 18\%$$

$$\text{Dry unit weight, } \gamma_{d2} = 1.76 \text{ g/cc}$$

Calculate Dry Unit Weight of Borrow Soil

$$\gamma_{d1} = \frac{G \gamma_w}{1 + e}$$

Taking

$$\gamma_w = 1 \text{ g/cc}$$

$$\gamma_{d1} = \frac{2.70 \times 1}{1 + 0.60}$$

$$\gamma_{d1} = 1.6875 \text{ g/cc}$$

Calculate Dry Mass Required for Embankment

$$\text{Dry Mass in Embankment} = \gamma_{d2} V_e$$

$$= 1.76 \times 40,000$$

$$= 70,400$$

Determine Borrow Pit Volume Required

$$V_b = \frac{\gamma_{d2} V_e}{\gamma_{d1}}$$

$$V_b = \frac{1.76 \times 40,000}{1.6875}$$

$$V_b = 41,718.52 \text{ m}^3$$

$$V_b \approx 41,720 \text{ m}^3$$

Quantity of soil to be excavated from the borrow pit = 41,720 m³.

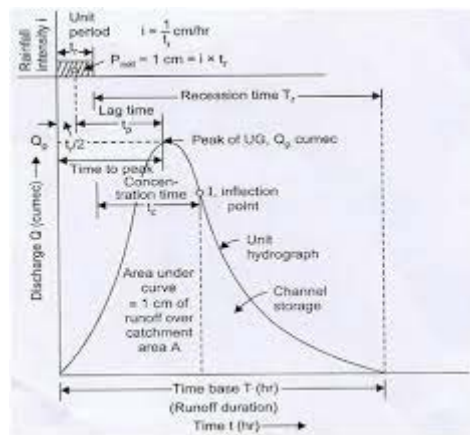
Section-B (25 Marks)

5. Define unit hydrograph. Explain its assumptions and limitations in flood forecasting for Nepalese rivers. [1+4 =5]

Answer:

Unit hydrograph is the direct run of hydrograph resulting from unit depth (1cm) of excess rainfall occurring at the uniform rate for the specific duration of D hours.

Unit hydrograph is the **unit pulse response function** of linear hydrologic system.



The basic assumptions of the unit hydrograph theory in flood forecasting in rivers of Nepal are listed below:

- The excess rainfall has constant intensity one by $\frac{1}{D}$ cm/hr for effective storm duration of D hours.
- The excess rainfall is uniformly distributed over the entire catchment area.
- The base time of direct runoff hydrograph (DRH) resulting from excess rainfall of given duration is constant.

The limitations of unit hydrograph theory are stated in the points below:

i. Rainfall Assumptions:

Unit hydrograph assumes the constant intensity of excess rainfall and uniform distribution of excess rainfall over entire catchment which is not possible in practicality as Nepal consists of varying landscape and rugged topography. The assumptions of uniform distribution of rainfall and constant intensity for application of UH fails to resemble the true characteristics of rainfall.

ii. Catchment Area Constraints:

Unit hydrograph is applicable for the catchment having area less than 5000km² square but should not be less than 2km². However, the theory doesn't talk about the large catchments having area greater than 5000 km².

iii. Other forms of precipitation:

It cannot be used if a major portion of the precipitation falls as snow or if a significant part of the basin is snow-covered. It is assumed that precipitation in the basin occurs only in the form of rainfall. However, in mountain regions of Nepal, where snow, glaze, hail are the major forms of precipitation is not considered by the UH theory.

iv. Large Natural Storage:

Catchments with large natural storages such as lakes, swamps, tanks or reservoirs alter the linear relationship between rainfall and runoff which makes the unit hydrograph inaccurate. Hence, this theory limits the presence of large natural storage in the catchment basin.

6. Introduce the concept of duty, delta and base period. Describe different methods of irrigation- surface, sprinkler and drip with the advantages, limitations and suitability in different agro-climatic zones of Nepal [4+6=10]

Answer:

Duty:

Duty of water is defined as the number of hectares that can be irrigated by constant supply of water at the rate of 1cumecs throughout the base period. The duty of water is not constant but it varies with various factors like soil conditions method of flowing method of application of water.

Delta:

If the total amount of water supplied to the crops from first watering to last watering is stored on land without any loss, then there will be a thick layer of water standing on the land. This depth of water is known as delta for the crop. It is denoted by Δ and expressed in cm.

Base period:

Base period is defined as the period from the first to last watering of the crop just before its maturity. It is denoted by B and expressed in number of days.

In open channel flow, the relationship between **duty (D)** and **delta (Δ)** is given by:

$$\Delta = \frac{8.64 \times B}{D}$$

or

$$D = \frac{8.64 \times B}{\Delta}$$

Where:

Δ (**Delta**) = Depth of water required by the crop during its base period (m)

D = Duty of water (hectares/cumec)

B = Base period of the crop (days)

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others. Few methods of irrigation are briefly described along with its advantages, limitations and suitability.

i. Surface irrigation:

- Surface irrigation is the most common type of the irrigation where the water is applied on the field in varied quantities and at different times.
- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Advantages:
 - a. It allows to store the required amount of water in the capillary zone of the soil for supply of root zone of plants.
 - b. It allows the use of machinery for land preparation cultivation and harvesting.
- Limitations:
 - a. Greater loss of water by surface runoff and deep percolation.
 - b. Inferior quality crops with a low yield are produced from the surface irrigation.
- Suitability:
 - Surface irrigation is most suitable for the land with the gentle slopes like terai region of Nepal.
 - This method is suitable for row and field crops.



ii. **Sprinkler irrigation:**

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- **Advantages:**
 - Provides protection to small plants from wind damage and soil from blowing.
 - Sprinklers provides protection against freezing injury to crops.
- **Disadvantages:**
 - Initial cost of installation is relatively high and regular maintenance must be done due to frequent clogging of the system.
 - This method is not so suitable at plain land as pressure head and sediment free water is required.
- **Suitability:**
 - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
 - It is suitable for closely spaced crops like cereals and fodders.



iii. **Drip irrigation:**

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.
- **Advantages:**
 - Reduced percolation and evaporation losses resulting in economical use of water.
 - Fertilizer and nutrient losses minimized due to localized application and reduced leaching.
- **Disadvantages:**
 - Frequent clogging of the dripper requires constant supervision.
 - Not suitable for very closely seeded crops such as carrots, radishes etc.
- **Suitability:**
 - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.

- The method is ideal for row crops, orchard and vineyards.



7. Describe the functions and components of a de-sanding basin in a hydropower project. Explain the design considerations for sediment flushing in the de-sanding basin. [5+5]

Answer:

De-sanding basin also known as de-stilling basin is a hydraulic structure placed between Intake and main stock in the hydropower project for removal of suspended sediment particles from flowing water which cannot be trapped by the gravel trap.

Functions of de-sanding basin:

- i. **Protection of turbine:**
De-sanding basin ensures that the water entering the conveyance system is free of sediment particles so that, wear and damage on the turbine blades and other part of the turbines are prevented.
- ii. **Increasing the efficiency of hydropower project:**
De- sanding basin increases the efficiency of the hydropower project by minimizing the damage On the structural component of the hydropower project.
- iii. **Consistent power generation:**
Descending basin also encourages the consistent power generation as it reduces maintenance requirements and need for frequent replacement of damaged parts of the project.

Components of de-sanding basin:

De-sanding basin consists of inlet, settling and outlet zones.

i. Inlet zone:

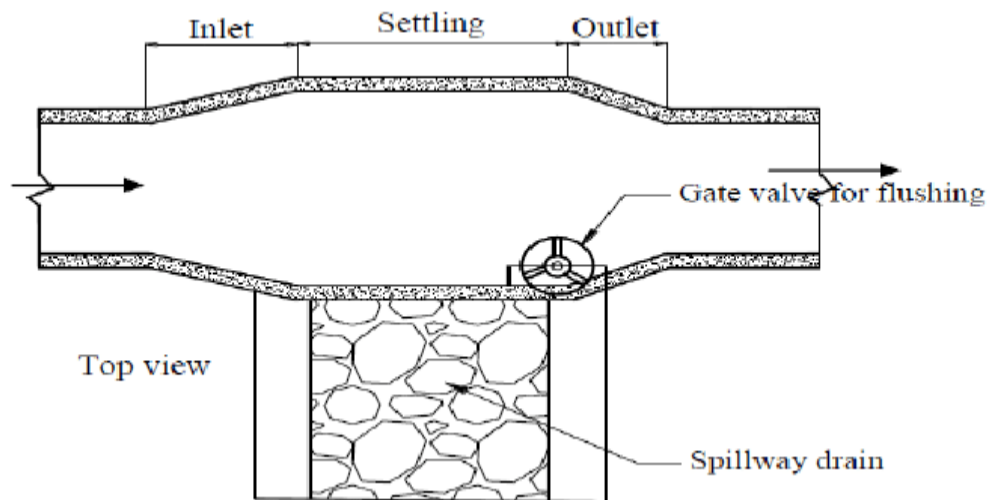
- Inlet zone consists of a transition designed for reducing the velocity of flow by gradually increasing the depth of basin.
- This allows an even floor distribution at the beginning of settling zone.
- The vertical expansion ratio is about 1:2 and Gradual expansion of inlet channel is about 1:5.

ii. Settling zone:

- It is the main part of basin where settling of suspended sediment takes place.
- The length of settling zone is also called effective length of basin.
- The provision for flushing of the stored sediment should be at the end of the settling basin
- Gentle slope of 1:2 to 1:5 is provided on the floor of settling zone to facilitate flushing.

iii. **Outlet zone:**

- Outlet zone is also provided with the transition to increase the velocity of flow into the conveyance system by gradually narrowing the width and depth of the basin.



The design considerations for sediment flushing in the de-sanding basin are listed below:

i. **Flushing velocity:**

- The flushing velocity must be high enough for the transportation of settled sediment out of settling basin.
- Usually, flushing velocity of 2-3 m/s must be maintained during flushing.

ii. **Flushing channel:**

- Flushing channels must be provided at the bottom of settling basin for the efficient collection and removal of sediments.

iii. **Bottom slopes:**

- Adequate longitudinal and transverse slope are provided to facilitate efficient movement of sediments towards flushing outlet.

iv. **Flushing gates:**

- Flushing gates should be located at the suitable positions and designed to pass sediment-laden water quickly and efficiently.

v. **Water requirement:**

- Sufficient water discharge is required to generate required flushing velocity.

Section-C (25 Marks)

8. Explain the factors governing the alignment of roads in hilly terrain of Nepal. [5]

Answer:

Hilly region of Nepal covers the mountainous & steep terrain having a cross slope of 25 to 60% and more. The road passing through the hilly terrain with a cross slope of 25% or more is generally termed as Hill Road. A hill road usually consists of either a river route or a ridge route. Road alignment in Nepal's hilly terrain is governed by a combination of **geological, topographic, drainage, economic, and geometric design factors**. The steep,

unstable slopes and complex geology of the Himalayas and Middle Hills make the design and construction of hill roads more critical.



i. Geological Stability:

- Road alignment should pass through stable hill slopes and should avoid landslide and rockfall prone zones.
- Construction activities might trigger landslide and rockfalls if the road alignment is to be passed through the area having unstable or weak hill slopes.
- Excessive cutting works through hard solid rocks during construction must be avoided.

ii. Cross drainage structures:

- Hilly region consists of large number of fast flowing, steep rivers that requires numerous cross-drainage works.
- The road alignment should be selected such that the roadway consists of minimum number of cross drainage structures.

iii. Geometric Design:

- Geometric design of hill roads must be done according to the design guidelines and standards for hill region.
- The alignment should have minimum number of sharp curves and hairpin bends, gentler gradient.

iv. Availability of construction materials:

- The road alignment should be selected in such a way that the construction materials are available near the construction site.
- It reduces the transportation cost of materials and makes the construction more economical.

v. Environment and weather:

- Rainfall (or Snowfall) $\propto$ Altitude
- As the altitude decreases, the number of cross-drainage works required increases.
- At higher altitudes, the road pavement may witness snowfall during winter. This is why the alignment of hill roads should preferably be provided at an altitude between 900 m slopes exposed to sun.

vi. Length:

- The cost of construction of a hill road per kilometer length is comparatively very high.

- Hence, it should be ensured that the length of the road connecting two stations should be minimum possible.

9. The speed of overtaking and overtaken vehicles are 70 and 40 kilometer per hour respectively on a two-way traffic. If the acceleration of overtaking vehicle is 0.99 m/s^2 .

a. Calculate the safe overtaking distance.

b. Mention the minimum length of overtaking zone with a neat scale of overtaking zone showing position of signpost. [4+6=10]

Answer:

Given,

Speed of overtaking vehicle, $V = 70 \text{ km/hr}$

Speed of overtaken vehicle, $v = 40 \text{ km/hr}$

Acceleration of overtaking vehicle, $a = 0.99 \text{ m/s}^2$

Reaction time = 2.5 sec

$$V = \frac{70 \times 1000}{3600} = 19.46 \text{ m/s}$$

$$v = \frac{40 \times 1000}{3600} = 11.11 \text{ m/s}$$

Distance Traveled by Overtaking Vehicle:

$$d_1 = vt$$

$$d_1 = (11.11)(2.5)$$

$$d_1 = 27.775 \text{ m}$$

Spacing between the vehicles:

$$s = 0.69 v + 6.1$$

$$s = 0.698 * 11.11 + 6.1$$

$$s = 13.86 \text{ m}$$

Time Required for Overtaking

spacing and reaction time is:

$$T = \sqrt{\frac{4 * S}{a}}$$

$$T = \sqrt{\frac{4 * 13.76}{0.99}}$$

$$T = 7.48 \text{ s}$$

Distance

$$d_2 = 2s + v * T$$

$$d_2 = 2 * 13.86 + 11.11 * 7.48$$

$$d_2 = 110.89 \text{ m}$$

Traveled by Opposing Vehicle

$$d_3 = Vt$$

$$d_3 = (19.46)(7.48)$$

$$d_3 = 145.56 \text{ m}$$

Safe Overtaking Distance

$$OSD = d_1 + d_2 + d_3$$

$$OSD = 27.775 + 110.89 + 145.56$$

$$OSD = 284.225 \text{ m}$$

$$OSD \approx 285 \text{ m}$$

(b) Minimum Length of Overtaking Zone

According to highway design standards:

$$\begin{aligned}\text{Minimum Length of Overtaking Zone} &= 3 \times OSD \\ &= 3 \times 285 \\ &= \boxed{855 \text{ m}}\end{aligned}$$

Desirable Length

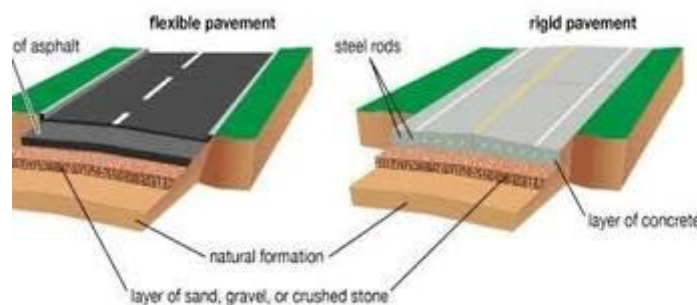
$$\begin{aligned}&= 5 \times OSD \\ &= 5 \times 285 \\ &= \boxed{1425 \text{ m}}\end{aligned}$$

10. Explain in brief the difference between flexible and rigid pavements and the suitability over each other. Write down the step-wise procedure of constructing asphalt concrete pavement in short. [5+5=10]

Answer:

A relatively stable layer constructed over the natural soil for supporting and distributing the wheel loads and provide adequate surface for movement of vehicles with safety, comfortability and economically is known as road pavements. Based upon the structural behavior of the materials used in construction, the pavements can be divided as:

- i. flexible pavement
- ii. rigid pavement
- iii. semi rigid pavement and
- iv. composite pavements.



The differences flexible and rigid pavements and the suitability over each other are as follows:

Flexible pavements	Rigid pavements
1. Flexible pavements on the heavy load yield to excessive stresses resulting in local depression of the surface.	1. Rigid payment under a heavy load rupture, producing a crack to the surface.
2. Flexible pavement with subgrade of varying thickness will adjust itself to irregularities due to differential settlements.	2. Rigid pavement with the subgrade of varying strength will not adjust to irregularities.
3. Temperature variations due to atmospheric conditions do not produce stress.	3. Temperature variations produce heavy temperature stresses.
4. Flexible pavement has self-healing properties.	4. Rigid pavement do not have self-healing properties.
5. Strength of flexible layer is by distributing the load over subgrade.	5. Strength of rigid layer is by bending action.

Suitability of flexible pavements:

- i. Lower initial construction cost with high suitability in expanding road networks.
- ii. Highways opened for operation of traffic immediately after its construction but rigid pavements require curing for 14+ days.
- iii. Easy maintenance works such as localized patchworks.
- iv. Flexible pavements can adapt to minor subgrade settlement and has healing properties.

Suitability of Rigid pavements:

- i. Best suited for the roads subjected to heavy and repetitive traffic loads such as industrial roads.
- ii. Requires less maintenances than flexible pavements.
- iii. Longer design lifespan making it economical.

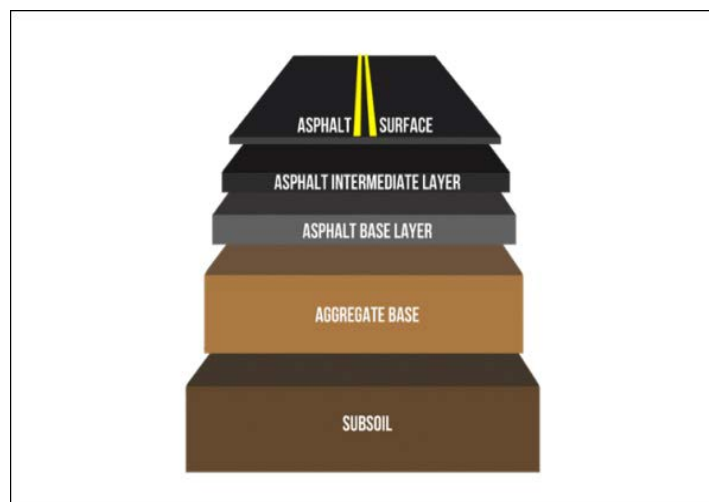
Step-wise procedure of constructing asphalt concrete pavement:

Asphalt concrete pavement (also called asphalt concrete, blacktop, or simply asphalt) is a composite material used to construct roads that consists of mineral aggregates bind together with a bitumen binder, creating a strong, flexible, and durable paving surface.



An asphalt concrete pavement is built as a **layered system** that distributes traffic loads from the surface to the ground. The structure typically includes **four main layers**:

- a. Subgrade soil: Existing ground surface
- b. Sub base: thickness varies from 4-12 inches.
- c. Base course: Aggregated stone or asphalt-treated base (stone + asphalt binder) with thickness 4-8 inches.
- d. Asphalt concrete layers: Aggregates + asphalt binder (bitumen)



Procedure of constructing asphalt pavement:

- i. **Preparation of subgrade:**
Subgrade soil is properly cleaned, leveled and compacted to provide required gradient and camber to the road pavement.
- ii. **Construction of subbase:**
Lay the granular subbase material providing the designed thickness and compact it properly.
- iii. **Construction of base course:**
Spread the base course material over the subbase course and compact it providing the required thickness.
- iv. **Application of prime coat:**
A layer of prime coat is applied above the base course to bind the loose particles for better bonding and performance of road pavement.
- v. **Application of tack coat:**
A layer of tack coat is sprayed at the existing top surface before laying asphalt mixes to improve proper adhesion.
- vi. **Preparation and transportation of asphalt mix:**
Prepare the asphalt mix in a hot mix plant with the suitable ingredients and transport it to the site.
- vii. **Laying of asphalt concrete:**
Lay the hot mix of asphalt mix uniformly over the surface by using paver machine.
- viii. **Compaction:**
Proper compaction of the layer must be done by the help of rollers to achieve required dry density.
- ix. **Quality control:**
Thorough checks regarding the thickness of pavements, density, smoothness must be carried out before finishing works. Rectify if any defects found.
- x. **Finishing and opening of traffic:**
Proper finishing works are carried out and then opened for traffic after giving the pavements enough time to cool.

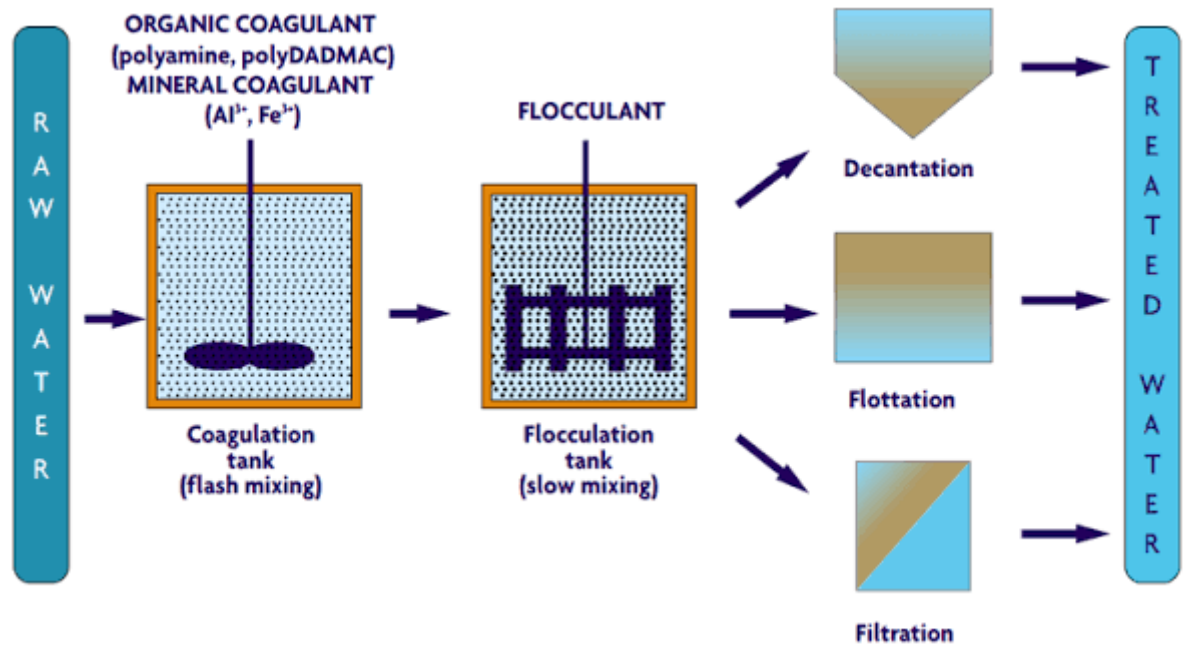
Section-D (25 Marks)

11. Describe the coagulation-flocculation process in water treatment. Differentiate between slow sand and rapid sand filters. Explain the cleaning mechanism of rapid sand filters. [4+6=10]

Answer:

- Coagulation-flocculation process in water treatment is a chemical and physical water treatment process used to remove suspended solids, turbidity, and color.
- Coagulation is a chemical process in which colloidal particles are brought together by addition of coagulants to water and mixing it thoroughly.
- Coagulants are the chemicals when added to water makes insoluble, gelatinous, flocculent precipitation that are removed by settling the suspended and colloidal particles from raw water by process of sedimentation.

- Most common coagulant used for coagulation are: alum/aluminum sulphate, iron salts, sodium aluminate etc.
- Flocculation is the process of formation of floc. The zones of floc are positively charged and they react with negatively charged particles to form heavier floc, which settles down under gravitational force.
- The main purpose of coagulation-flocculation process is to remove very fine suspended and colloidal particles and reduce load in subsequent treatment plants.



Filtration:

The process of passing the water through beds of sand for removing small suspended sediments present in the water that cannot be removed by sedimentation is known as filtration. On the basis of filtration rate, filters are classified into three types. They are:

- Slow sand filters
- Rapid sand filters
- Pressure filters

The differences between slow sand filter and rapid sand filter are stated in points below:

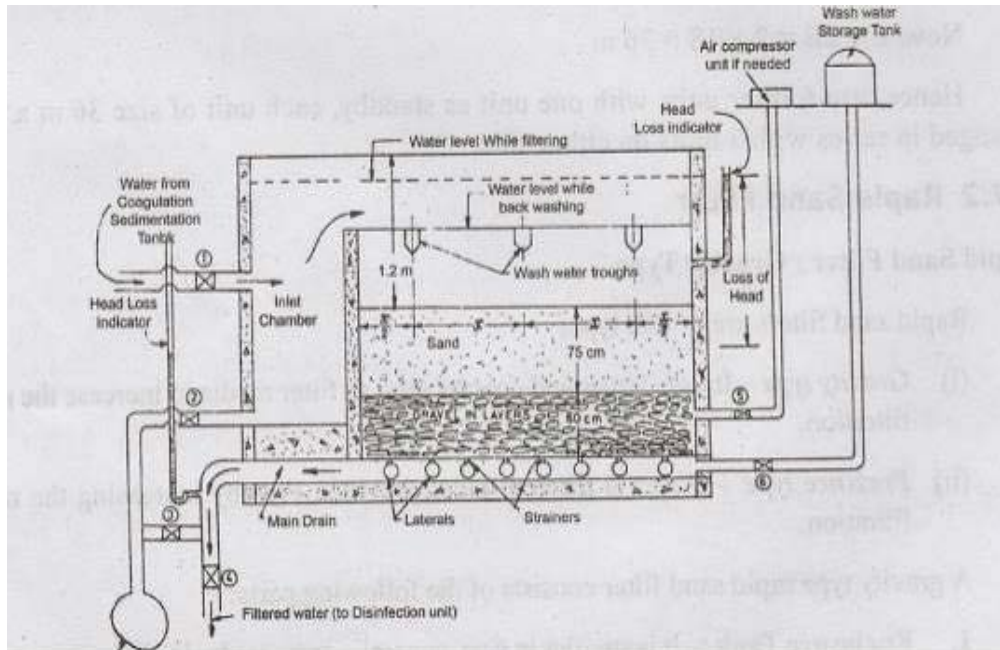
Slow sand filter	Rapid sand filter
1. This type of filters has very slow rate of filtration hence called as slow sand filters.	1. This type of filters has high rate of filtration hence called as rapid sand filters.
2. Uneconomical and costly to install.	2. Economical and less costly to install.
3. Slow sand filter requires large area and large quantity of sand.	3. Rapid sand filter requires less area and small quantity of sand.
4. Water having low turbidity (less than 20 NTU) can be directly feed to slow sand filter without any pretreatment.	4. Pretreatment of raw water is required before passing the water through rapid sand filter.
5. Backwashing is not provided in slow sand filter.	5. Backwashing is provided in rapid sand filter which is done in every 1-3 days.
6. slow sand filter is highly efficient in removal of bacterial load.	6. Rapid sand filter is highly efficient in color removal bus less effective to remove bacterial load.

Working and cleaning of RSF:

During the normal stage of RSF only valve 1 and 2 are opened and rest of the other valves are kept closed. Valve 1 permits entrance of water from sedimentation with coagulation and valve 2 allows the filtered water out, from filter to clear water reservoir. The level of water over the sand bed is kept between 1-2 m.

Backwashing is usually done when head loss is reached between 2.5 to 3 m or filter cleaning is done at an interval of 1-3 days up to about 30 minutes.

For backwashing:



- first close valve 1 and as the level reaches the edge of wash trough close valve 2.
- Now, valve 6 is opened to permit compressed air about 2-3 minutes in upward direction surface. Scum will break-up and loosen the dirt.
- Then, Valve 6 is closed and valve 4 is opened to allow wash water, valve 3 is opened to drain dirty wash water till clear water is obtained.
- Now, valve 3 and 4 is closed and allow certain time to settle the materials.
- Then valve 1 is opened slightly and valve 5 is opened for few minutes to discharge the filtered water to wash the drain.
- Finally, valve 5 is closed and valve 2 is opened. New filter can be operated at normal filtration rate.

12. Mention different types of environmental studies required before any construction activities as per existing EPA 2076 and EPR 2077. Write down stepwise procedure for EIA study and its approval. [4+6=10]

Answer:

As per EPA 2076 and EPR 2077, there are three types of environmental studies required before any construction activities. They are:

i. Basic Environmental Study (BES):

- According to the Environment Protection Act (EPA) 2076, Basic Environmental Study (BES) is a brief environmental assessment conducted to identify the possible environmental impacts of a proposed project and to recommend measures to prevent or minimize those impacts.
- It is required for projects that are likely to have only minor environmental effects.
- Projects requiring BES are listed in **Schedule 1 of EPR 2077**.
- BES is done for
 1. length of road less than 5 km
 2. irrigation project with command area less than 25 hectares
 3. hydropower project with installed capacity of less than 1 MW and so on.

ii. Initial Environment examination (IEE):

- According to Environment Protection Act (EPA) 2076 and Environment Protection Rules (EPR) 2077, IEE is defined as "An analytical study or evaluation conducted to determine whether a proposed project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts."
- Done for medium scale projects that has moderate effect on environment.
- Project requiring IEE are listed in **Schedule 2 of EPR 2077**.
- Includes:
 - Screening → Scoping → Preparation and approval of work schedule → Field study and public hearing → IEE Report Preparation → Recommendation and submission → final review and approval → Decision making
- If IEE Report submitted is approved by concerned authority, the project is implemented.
- IEE is done for:
 - a. Highways with the length of road 5-50 km
 - b. Irrigation project with command area 25-2000 hectares.
 - c. Hydropower project with installed capacity of 1 MW-50 MW

iii. Environmental Impact assessment (EIA):

- "Environmental impact assessment" is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.
- Projects requiring EIA are listed in **schedule 3 of EPR 2077**.
- Includes:
 - Screening → Scoping → Preparation and approval of work schedule → Field study and public hearing → EIA Report Preparation → Recommendation and submission → final review and approval → Decision making
- EIA is required for large-scale development projects with significant environmental impacts that cannot be adequately assessed through a Brief Environmental Study (BES) or Initial Environmental Examination (IEE).
- EIA is done for:
 - a. Highways with the length of road greater than 50 km
 - b. Irrigation project with command area greater than 2000 hectares.
 - c. Hydropower projects with installed capacity of greater than 50 MW.

iv. Strategic Environmental Analysis (SEA):

- A **Strategic Environmental Analysis** is required for high-level policies, programs, or large-scale projects specified by the Government of Nepal through notice in the Nepal Gazette.
- Unlike project-specific studies (BES/IEE/EIA), an SEA evaluates the environmental implications of broader strategies and plans before they are implemented to ensure sustainable development at a macro level.

Stepwise procedure for EIA study and its approval:

i. Screening:

- This step determines whether the proposed project requires IEE or EIA.
- No project requiring EIA can be implemented without approval of the environmental study report.

ii. Preparation of Scoping Document:

The proponent prepares Scoping Document following the format in **Schedule-5** and is submitted to the concerned approving authority.

- For federal projects, it is submitted to the **Concerned Ministry** (e.g., the Ministry of Irrigation or Physical Infrastructure).
- The Concerned Ministry reviews it, adds its opinion, and forwards it to the **Ministry of Forests and Environment (MoFE)** within **15 days**.

- MoFE must approve or modify the Scoping Report within **15 days** of receiving the recommendation.

iii. Preparation and Approval of work schedule:

- The proponent prepares a **Work Schedule** (per **Schedule-8**) that outlines the methodology and timeline for the study.
- The Scoping Report and Work Schedule are often submitted and approved together.
- Once reviewed, MoFE must approve the Work Schedule within **15 days**.

iv. Field study and Public Hearing:

- The study must be carried out by a team of qualified experts who meet the academic and experience criteria listed in **Schedule-13**.
- The proponent **must organize at least one public hearing** in the project area to gather feedback from local communities, forest user groups, and local representatives.
- The proponent **must publicize the hearing** via local radio, newspapers, or other media, and post notices at the ward office. Documentation of the hearing (attendance, suggestions, photos, and audio-visual recordings) must be included in the final report.

v. EIA Report Preparation:

- EIA report is prepared based on the approved Scoping Report, Work Schedule, and the suggestions from the public hearing according to the detailed format in **Schedule-12**.
- **Public Notice for Feedback:** The proponent must issue another **7-day notice** (Schedule-9) in a national daily newspaper to gather written suggestions on the potential environmental impacts of the project.
- Reports are generally prepared in **Nepali** language.
- For foreign-invested projects, they may be in English, provided a concise **executive summary in Nepali** is included.

vi. Recommendation and Submission:

- The proponent must include a formal **Recommendation Letter (सिफारिस पत्र)** from the concerned **Local Level** and the relevant thematic office (per **Schedule-14**). These bodies must provide the recommendation within **15 days**.
- **Submission:** For federal projects, the report is submitted to the **Concerned Ministry**.
- The Concerned Ministry reviews the report and forwards it to **MoFE** with its formal opinion within **15 days**.
- The final report must be submitted within **two years** of the Scoping/Work Schedule approval.

vii. Final Review and Approval of EIA:

- **Final Public Review:** Upon receiving the EIA report, MoFE must make it public for **7 days** on its website and via a national daily newspaper to collect final public comments.
- **Review Committee:** MoFE may form a **committee** of representatives and experts to investigate the report, which may include on-site inspections or monitoring.
- **Approval Decision:** If the investigation shows the project will not cause significant adverse environmental impacts, MoFE shall approve the EIA report within **35 days**. The approval may include specific, binding conditions that the proponent must follow.
- Additionally, the project must start within **three years** of approval, or the proponent may be required to conduct a new study.

Note on Violations: Implementing a project without an approved EIA report or in violation of its conditions can lead to a fine of up to **5 million (५० लाख) rupees**.